Chapter 9 — Electrostatics and Current Electricity

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Coulomb force: (a) qr/k (b) $kq_1q_2r^2$ (c) kq_1/q_2r (d) kq_1q_2/r^2

2. Electric field is: (a) q/F (b) Fq (c) $F + q$ (d) F/q

3. Field between plates: (a) d/V (b) V/d (c) Vd (d) $V + d$

4. One electron volt: (a) 1 C (b) 1.6×10^{-19} C (c) 1 V (d) 1.6×10^{-19} J

5. Ohm's law: (a) $V = R/I$ (b) $V = I + R$ (c) $V = I/R$ (d) $V = IR$

6. Resistance of wire: (a) $\rho A/L$ (b) LA/ρ (c) $\rho L/A$ (d) ρ/LA

7. Terminal voltage: (a) E/Ir (b) $Ir - E$ (c) $E - Ir$ (d) $E + Ir$

8. LDR in light: (a) resistance falls (b) becomes zero (c) unchanged (d) resistance rises

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why does the electric force decrease rapidly with distance?

2. Why is the electric field inside a charged conductor zero?

3. Why does a bulb light immediately although electron drift is slow?

4. Why is terminal voltage less than emf when current flows?

5. Why does the galvanometer resistance not affect a balanced Wheatstone bridge?

6. How does an LDR work in a potential divider?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain emf and internal resistance and derive $V = E - Ir$. **(4)**

(b) State Kirchhoff's rules and explain the sign conventions for loop analysis. **(4)**

Q.4 (a) Explain the Wheatstone bridge and derive its balance condition. **(4)**

(b) Describe the potentiometer and explain how thermistors and LDRs are used as sensors. **(4)**

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